

Make It Do Something: Dynamic Contact & Ingestion Feature

Assignment: Week 6 (General AI Fluency) | Author: Muhammad Akbar Pradana (ML Intern @ FlyRank AI)

Target System: akbarprdna.my.id / akbarpradana.netlify.app

1. Feature Selection & Purpose

A static portfolio operates merely as a digital poster. To transform the portfolio into an active, functional tool, I implemented a **Serverless Contact & Project Collaboration Pipeline** operating on a zero-cost tier (Formspree / Netlify Forms API). This enables hiring managers, recruiters, and collaborators to submit direct inquiries that trigger automated real-time email routing to my primary inbox with zero server maintenance overhead.

2. Plain-Words Explanation: What is a Backend?

In web architecture, the **Frontend** is what the user sees and interacts with in their web browser (the visual layout, text, buttons, and input fields built with HTML, CSS, and JavaScript).

The **Backend** is the invisible engine running on a remote cloud server that performs the heavy lifting the browser cannot do alone: securely handling credentials, validating user inputs to prevent spam, and interfacing with third-party networks (e.g., SMTP email servers or relational databases) to permanently record and dispatch data.

3. End-to-End Data Flow Architecture

The data lifecycle follows a 5-step asynchronous request-response cycle:

Step	Component	Technical Operation
1. Input	Browser Client	User enters name, email, and inquiry details into the HTML form.
2. Dispatch	HTTP POST (JSON)	JavaScript intercepts submit, serializes JSON payload, and posts to API endpoint.
3. Backend	Serverless Gateway	Cloud function validates email syntax, rate limits, and checks anti-bot honeypot.
4. Routing	SMTP Dispatch	Backend triggers automated email forward directly into primary inbox.
5. Feedback	HTTP 200 Response	Browser receives confirmation status and renders success banner to user.

4. Verification & Testing Proof

- **Live Test Run:** Dispatched verified test payload from portfolio interface.
- **Delivery Confirmation:** Successfully received formatted email alert with correct timestamp, client headers, and sender details.
- **Error Handling:** Gracefully catches invalid input patterns and offline network states.